

Assuming u_1, u_2 are solutions of the equation, let $u = u_1 - u_2$:

$$\begin{cases} \frac{d}{dt}u + vAu = B(u_2) - B(u_1), & u \in L^\infty(0, T; V) \cap L^2(0, T, D(A)) \\ u(0) = 0 \end{cases}$$

By lemma 2.8:

$$\frac{1}{2} \frac{d}{dt} \|u_m\|_{L^2}^2 + va(u, u) = (B(u_2) - B(u_1), u_2 - u_1)$$

Then we observe:

$$B(u_1) - B(u_2) = B(u_1, u) + B(u, u_2)$$

and

$$\begin{cases} (B(u_1, u), u) = 0 \\ |(B(u, u_2), u)| = \left| \int_{\Omega} u \cdot \nabla u_2 \cdot u dx dy \right| \leq \|u\|_{L^\infty} \|\nabla u_2\|_V \|u\|_{L^2} \end{cases}$$

Due to $H^1(\Omega) \hookrightarrow L^\infty(\Omega)$, with Agmon's inequality, $|(B(u, u_2), u)| \leq c \|u\|_{L^2}^{3/2} \|u_2\|_V \|u\|_V^{1/2}$. By Young inequality, $|(B(u, u_2), u)| \leq \frac{v}{2} \|u\|_V^2 + \frac{c^2}{2v} \|u\|_H^2 \|u_2\|_V^2$. Hence,

$$\frac{d}{dt} \|u\|_H^2 \leq \frac{c^2}{v} \|u\|_H^2 \|u_2\|_V^2$$

By Gronwall inequality, we obtain:

$$\|u(t)\|_H^2 \leq \|u(0)\|_H^2 e^{\frac{c^2}{v} \int_0^t \|u_2(s)\|_V^2 ds} = 0$$

Hence, the solution is unique. And the continuous dependence of the solution on f and the initial value u_0 can be deduced by the prior estimation.

3 Application

The Burgers equation, originally derived as a simplified model of fluid dynamics, has found widespread applications in various fields, including statistical mechanics, turbulence modeling, and transportation systems. Its nonlinearity and ability to describe shock formation and energy dissipation make it a powerful tool for analyzing complex systems. In transportation, the Burgers equation is particularly useful for modeling traffic flow dynamics, where it helps capture phenomena such as congestion, wave propagation, and the formation of traffic jams.

3.1 Introduction of Detailed Application in Transportation Systems

The application of the Burgers equation in transportation systems has been extensively studied. For instance, Payne (1971, 1979) utilized the Burgers equation to develop mathematical models for traffic flow, focusing on the propagation of shock waves and the behavior of traffic under varying conditions. These models provided insights into the dynamics of traffic congestion and helped improve traffic management strategies.

Kühne (1984) further expanded on this work by applying the Burgers equation to simulate traffic flow in urban networks. His studies demonstrated how the equation could be used to predict the formation and dissipation of traffic jams, offering valuable tools for urban planning and traffic control.

The Burgers equation's ability to describe the nonlinear interactions between vehicles and the environment makes it a cornerstone in traffic flow theory. By incorporating viscosity and external forcing terms, researchers can model the effects of road conditions, driver behavior, and traffic regulations on flow dynamics.